Solution

Since $v(3)=64,\ v(6)=133,$ and $v(9)=208$, we get the following system of linear equations

$$\begin{aligned} 9a+3b+c &= 64, \\ 36a+6b+c &= 133, \\ 81a+9b+c &= 208. \end{aligned}$$

We solve the above system of linear equations by Gaussian elimination method.

Reducing the augmented matrix to an equivalent row-echelon form by using elementary row operations, we get

$$[A\mid B]=\left[\begin{array}{ccc|c} 9 & 3 & 1 & 64 \\ 36 & 6 & 1 & 133 \\ 81 & 9 & 1 & 208 \end{array}\right] \xrightarrow{R_2\to R_2-4R_1,\ R_3\to R_3-9R_1} \left[\begin{array}{ccc|c} 9 & 3 & 1 & 64 \\ 0 & -6 & -3 & -123 \\ 0 & -18 & -8 & -368 \end{array}\right] \xrightarrow{R_2\to R_2\div(-3),\ R_3\to R_3\div(-2)} \left[\begin{array}{ccc|c} 9 & 3 & 1 & 64 \\ 0 & 2 & 1 & 41 \\ 0 & 9 & 4 & 184 \end{array}\right]$$

$$\xrightarrow{R_3\to 2R_3} \left[\begin{array}{ccc|c} 9 & 3 & 1 & 64 \\ 0 & 2 & 1 & 41 \\ 0 & 18 & 8 & 368 \end{array}\right] \xrightarrow{R_3\to R_3-9R_2} \left[\begin{array}{ccc|c} 9 & 3 & 1 & 64 \\ 0 & 2 & 1 & 41 \\ 0 & 0 & -1 & -1 \end{array}\right] \xrightarrow{R_3\to(-1)R_3} \left[\begin{array}{ccc|c} 9 & 3 & 1 & 64 \\ 0 & 2 & 1 & 41 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

Writing the equivalent equations from the row-echelon matrix, we get

$9a+3b+c=64,\ 2b+c=41,\ c=1.$

By back substitution, we get $c=1,\ b=\dfrac{(41-c)}{2}=\dfrac{(41-1)}{2}=20,\ a=\dfrac{64-3b-c}{9}=\dfrac{64-60-1}{9}=\dfrac{1}{3}.$

So, we get $v(t)=\dfrac{1}{3}t^2+20t+1.$ Hence, $v(15)=\dfrac{1}{3}(225)+20(15)+1=75+300+1=376.$ ■

EXERCISE 1.5

1. Solve the following systems of linear equations by Gaussian elimination method:

   (i) $2x-2y+3z=2,\quad x+2y-z=3,\quad 3x-y+2z=1$

   (ii) $2x+4y+6z=22,\quad 3x+8y+5z=27,\quad -x+y+2z=2$

2. If ax^2+bx+c is divided by $x+3, x-5,$ and $x-1,$ the remainders are $21, 61$ and 9 respectively. Find a, b and c. (Use Gaussian elimination method.)

3. An amount of ₹ 65,000 is invested in three bonds at the rates of $6\%, 8\%$ and 9% per annum respectively. The total annual income is ₹ 4,800. The income from the third bond is ₹ 600 more than that from the second bond. Determine the price of each bond. (Use Gaussian elimination method.)

4. A boy is walking along the path $y=ax^2+bx+c$ through the points $(-6,8), (-2,-12)$, and $(3,8)$. He wants to meet his friend at $P(7,60)$. Will he meet his friend? (Use Gaussian elimination method.)

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